Applying Engineering Economics – Project Reflection

Course Learning Outcomes that will be focused upon for the reflection assignment

1. Utilize economic principles to make decisions in engineering projects.
2. Make reasonable assumptions or perform necessary research to cope with ambiguity and uncertainty in required tasks.
3. Apply the fundamentals of cost, price, present value, and other financial metrics.
4. Manage group projects and interpersonal relations, specifically building project management skills, preparing agendas, and diagnosing team dynamics.
5. Professionally communicate complex decisions following an economic analysis.

Write a reflection that addresses each of the prompts below. Please consider the Main Project in your reflection. (Length 400 to 600 words + ILO Survey)

Description

- What was your Main Project about?
- Who did you work with? What were your contributions?
- What were your group members' contributions?

Feelings

- What did you hope to gain from this project?
- Did you feel equipped to take on this project? Why or why not?
- Do you feel the Design Studios and lectures help you take on this project? Why or why not?

Evaluation

- How did it go? What was your initial reaction to the experience?
- Did this experience meet your expectations? Why or why not?
- What did you think went well?
- How did the Design Studios and lectures help you prepare for the Main Project?

Analysis

- What past experiences helped equip you for this experience?
- Regarding the Intended Learning Outcomes (ILO) of the course, how did this experience bring you closer to achieving these outcomes?

- Is there an ILO that this experience helped you achieve? Which one? Why or why not?
- Is there a particular ILO that this experience did not help as much as you expected? Which one? Why or why not?

Conclusion

- What skills/techniques employed during this experience did you find valuable and useful for future experiences (either similar or different to this project)?
- If you had to work on this project again, what would you do differently or similarly?

Group_Section_Week12_SelfReflection

Please enter your response here.

Description

The Main Project required convincing key stakeholders at a hypothetical Canadian Space Agency facility that our proposed Facility Security Software was economically viable over its service lifespan. I had to demonstrate that both Present Worth (PW) and Annual Worth (AW) exceeded zero even under worst-case conditions, and that the Internal Rate of Return (IRR) surpassed the Minimum Acceptable Rate of Return (MARR). Jacob and I handled the Software subsystem, while other three teammates independently developed the Electrical Surveillance subsystem.

Within my subsystem, I independently identified the problem statement, defined key stakeholders, mapped the critical path, determined design parameters, and computed net cash flow, AW, and PW. I also collaborated with all members to establish shared assumptions. My design was ultimately not selected (Jacob's yielded superior results in terms of net cash flow, PW and AW) but this led to a productive collaboration. During Week 9 and 10, we co-developed the risk matrix, identified risk drivers, performed sensitivity analysis, and ran 500 Monte Carlo simulations. The PW remained positive across all runs, confirming the design's robustness.

Feelings

I entered this project hoping to evaluate a project's financial viability before any money is committed. Initially, concepts like discount rates, inflation adjustments, and Present Worth felt abstract. That uncertainty gave way to confidence through Jacob's guidance, TA support during Design Studios, and the instructor's Excel demonstrations. One suggestion I would offer is incorporating real-time, hands-on Excel practice during lectures, as I learn most effectively by doing rather than observing.

Evaluation

The project concluded smoothly with a constructive TA interview that consolidated my learning. I was genuinely surprised by how much I had grown; I now feel confident pursuing engineering project management roles and helping decision-makers evaluate projects economically.

The experience met my expectations, though Monte Carlo simulation proved more nuanced than anticipated. My initial attempt was incorrect because I had not defined stochastic terms, causing all 500 runs to return identical PW values. Jacob caught this error and corrected my approach. He also flagged my inconsistent lifespan assumption (10 years versus his 15), which would have invalidated any direct comparison. Design Studios were particularly valuable, they offered a structured setting to collaborate, consult TAs in real time, and refine understanding before each deliverable.

Analysis
My first-year Introductory Microeconomics course at McMaster provided a useful foundation. Familiarity with cost structures and price sensitivity gave me an intuitive starting point for financial modelling within this project.

This project directly advanced CLO 3 "Apply the fundamentals of cost, price, present value, and other financial metrics" through hands-on computation of net cash flow, PW, AW, and IRR, making abstract concepts concrete.

My Monte Carlo mistakes deepened my understanding of CLO 2 "Make reasonable assumptions to cope with ambiguity and uncertainty." Failing to define stochastic terms taught me that rigorous assumption-setting is foundational to credible analysis.

However, CLO 4 "Manage group projects and interpersonal relations" was not advanced as much as I expected. The project lacked structured practice like preparing agendas, and group skills developed reactively through resolving disagreements with Jacob rather than deliberate practice.

Conclusion
The most transferable skills from this project are risk identification, sensitivity analysis, and communicating financial data to non-technical stakeholders. Probabilistic thinking through Monte Carlo simulation, evaluating projects under stress, not just ideal conditions, is a mindset I will carry into product management and beyond.

If I repeated this project, I would push for a real client and facility, as our biggest limitation was the quality of estimated assumptions. I would preserve the collaborative dynamic with Jacob, as his accountability was among the strongest drivers of my learning.

ILO Survey:

To what extent have you achieved the Intended Learning Outcome of the course? (Mark an X)	Strongly disagree	Disagree	Neither agree nor disagree	Agree	Strongly agree
I can utilize economic principles to make decisions in engineering projects.					X
I can make reasonable assumptions or perform necessary research to cope with ambiguity and uncertainty in required tasks.				X	
I can apply the fundamentals of cost, price, present value, and other financial metrics.					X

I can manage group projects and interpersonal relations, especially building project management skills, preparing agendas, and diagnosing team dynamics.					X
I can professionally communicate complex engineering decisions following an economic analysis.					X

Self-Reflection Rubric

	Level 0	Level 1	Level 2	Level 3	Level 4
Description (X/4)	No description of the project is given.	The description of the project does not include a personal perspective.	The description of the project is thorough, but it is written in a poor, confusing, or vague manner.	The description of the project is detailed and well-written but missing some key elements like personal responsibilities, contributions, team members' contributions, or project goals.	The description of the project is thorough, well-written, and clear. It effectively includes personal responsibilities, individual contributions, contributions from others, and the overarching project goals.
Feelings (X/8)	No description of thoughts and feelings was given.	The description of thoughts and feelings is surface-level and does not contribute to personal reflection.	The description of thoughts and feelings is thorough, but it is written in a poor, confusing, or vague manner or does not contribute to personal reflection.	The description of thoughts and feelings is thorough, well written and contributes to personal reflection. The section does not consider expectations or earlier experiences in the course.	The description of thoughts and feelings is thorough, meaningful, well written and contributes to personal reflection. It includes feelings at the beginning of the project, and expectations and mentions the inputs of earlier experiences in the course.
Evaluation (X/16)	No evaluation of what was positive and negative was given.	Evaluation of what was positive and negative is surface level and does not contribute to personal reflection.	Evaluation of what was positive and negative about the experience is thorough, but it is written in a poor, confusing, or vague manner or does not contribute to personal reflection.	Evaluation of what was positive and negative about the experience is thorough, well written and contributes to personal reflection. The section does not consider expectations or aspects of the 3PX3 course that helped in preparing students.	Evaluation of what was positive and negative about the experience is thorough, meaningful, well written and contributes to personal reflection. It includes expectations, positive aspects, aspects that could be improved and aspects of the 3PX3 course that helped in preparing students.
Analysis (X/16)	No analysis of what was positive and negative was given.	The analysis is surface-level and does not contribute to personal reflection.	Analysis is meaningful and thoughtful, but it is written in a poor, confusing, or vague manner or does not contribute to personal reflection.	The analysis is meaningful and thoughtful, well written and contributes to personal reflection. The section does not consider past experiences or clear reference to the ILOs	Includes meaningful and thoughtful analysis from the student's experience, is well written and contributes to personal reflection. It includes students' past experiences and clear references to the ILOs, reflecting on those that helped the most and those that did not help as much as the student expected.
Conclusion (X/16)	No reflection on what students learned from the experience was given.	Reflection on what was learned is surface-level and does not contribute to personal reflection.	Reflection on what was learned is meaningful and thorough, but it is written in a poor, confusing, or vague manner or does not contribute to personal reflection.	Reflection on what was learned is meaningful and thorough, well written and contributes to personal reflection. The section does not consider valuable skills/techniques or what students would do similarly or differently in the future.	Meaningful reflection on what the student learned from the experience, is well written and contributes to personal reflection. It includes valuable skills/techniques for future activities, and what students would do similarly or differently.
Writing/ Grammar (X/16)	Awkward sentences, repetitive. Numerous errors make the	Many errors in grammar, spelling, and mechanics.	Most sentences are well constructed. Few errors in grammar, spelling, and mechanics.	Well-constructed sentences, few errors in grammar, spelling, and mechanics.	Well-constructed sentences, no errors in grammar, spelling, and mechanics.

	essay difficult to understand.				
Survey (X/4)	Did not complete the Survey.				Completed Survey.

List of Penalties:	Deduction
Late submission	-20% per day
More than XX words or less than YY words	-10% per 50 words